COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 1 : Session – 4 : SLO – 1 | Number system conversions
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5 MCQs

• 1. a — 11001 (25 = 16+8+1)
• 2. b — 4E (78 = 4×16 + 14)
• 3. c — Octal
• 4. b — 110110 (groups 110 110 = 6 6)
• 5. c — 3C (111100 = 0011 1100 = 3 C)

2 Two-Mark Questions

• 1. 10110 = 1×16 + 0×8 + 1×4 + 1×2 + 0×1 = 16 + 4 + 2 = 22 (decimal).
• 2. 3F hex: 3 = 0011, F = 1111 → 00111111 (binary) = 63 decimal.

1 Scenario-Based Long Answer

• Decimal 78 → Binary (repeated division by 2): 78/2=39 r0, 39/2=19 r1, 19/2=9 r1, 9/2=4 r1, 4/2=2 r0, 2/2=1 r0, 1/2=0 r1. Reading remainders upward: 1001110.
• Decimal 78 → Octal (division by 8): 78/8=9 r6, 9/8=1 r1, 1/8=0 r1 → 116 (octal).
• Decimal 78 → Hexadecimal (division by 16): 78/16=4 r14(E), 4/16=0 r4 → 4E.
• Exact values: 78■■ = 1001110■ = 116■ = 4E■■.

5 Match the Following

• Decimal to Binary — c) 2■ weight conversion (repeated division by 2)
• Decimal to Hexadecimal — b) Divide by 16
• Binary to Octal — a) Group digits in 3s
• Hexadecimal to Binary — d) Replace each digit with 4-bit binary
• Octal to Binary — e) Replace each digit with 3-bit binary

5 Fill in the Blanks

• 1. 1001110 2. base 8 3. 31 4. 4 5. 3C

1 Scenario-Based Small Activity

• Binary 1010110 to octal: pad to groups of three from the right → 001 010 110.
• 001 = 1, 010 = 2, 110 = 6. Final answer: 126 (octal).

5 True or False

• 1. TRUE (11001 = 25) 2. FALSE (7A = 1111010, not 1110101)
• 3. TRUE (57■ = 101 111) 4. TRUE 5. TRUE (1010110 = 126■)